PROF. DR.-ING. MARK SCHUTERA

FULL STACK HANDWERK

UNFINISHED LECTURE NOTES

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These notes are, by their very nature, unfinished, and they improve with every reader. If you spot an error, disagree with a framing, or want to add a source, a question, or a position, open a pull request at https://github.com/Quillstacks/LectureMaterial/tree/main/lecturenotes/notes_missingsemester. Every contribution is welcome.

2026-10-12 · merry damson Ente

Projektbericht

2026-10-12 · sneaky date Kauz

Real artists ship. - Steve Jobs

The Why

A lecture that teaches tools and workflows must also assess tools and workflows. Written exams test recall; a project report tests whether you can actually build and deploy something. Enter the Projektbericht.

Pick any block from the course and build a meaningful extension onto it. The extension should add functionality, improve reliability, or strengthen security in a way that goes beyond what the lecture covered. The choice is yours, but each extension must be unique across the class: submit your proposal for approval on a first come, first served basis with me. If your proposal strikes home with me, we can also discuss small full stack projects, which live outside the scope of the lecture material.

Submit a written report as PDF and a link to your GitHub repository. The report must cover:

- Which block you chose and what it covers in the course.
- What you built on top of it, which methods, tools and techstack you used, and why it adds value.
- Open questions and problems you encountered and how you resolved them.
- What works, what remains incomplete, and what would you do next.

The Projektbericht as in the Prüfungsordnung. The official frame for the Projektbericht is set by the DHBW Studien- und Prüfungsordnung Wirtschaft, available here:

"Die zu prüfende Person soll zeigen, dass sie in der Lage ist, Projekte oder Studien mit wissenschaftlicher oder praktischer Problemstellung selbstständig zu bearbeiten sowie deren Ergebnisse schriftlich zu dokumentieren. Der Projektbericht soll je zu prüfender Person in Modulen mit fünf bis sechs ECTS-LP zehn bis zwölf Seiten, (bei uns entsprechend 5 Seiten)

...

umfassen."

How to write the report. A practical guide for writing reports and theses lives here.

For deeper druidic knowledge on writing a thesis, consult here.

Evaluation and its Five Dimensions. Depth of extension asks how far the extension reaches beyond the lecture block, in functionality, originality, method or combination. Technical execution asks for a working repository, clean and reproducible, software engineering best practices and supportive of your method. Understanding and structure shows in methods and your writing, design choices justified, and trade-offs grasped and motivated, carried by a clear outline and a line of argument the reader can follow. Reflection and rigor names problems and their resolutions, states gaps and limitations of the contribution honestly, and makes next steps concrete. Writing and format is concise, structured, and well argued prose inside a correct formal frame: roughly five pages, an overview figure, a clear IMRD structure, and the reader is always in mind.

Looking for inspiration? Explore what tools modern teams actually use here or browse the ThoughtWorks Technology Radar here. Pick something that excites you; the best reports come from genuine curiosity.

Musterlösungen live here, an MCP layer that makes the web application agent accessible, and here, a quantized language model served from a Raspberry Pi behind a zero trust tunnel. Use them to calibrate scope and structure, then write your own.

